TRAFFIC SIGN RECOGNITION SYSTEM

BY

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OVERVIEW

- This project presents a traffic sign recognition system using image preprocessing, feature extraction, and classification techniques.
- Color and shape features were extracted, and two models (SVM and CNN) were trained.
- The system achieved around 91% accuracy, with SVM slightly outperforming CNN.

INTRODUCTION

- Traffic sign recognition is a key application in computer vision
- It is widely used in autonomous driving systems
- It helps improve safety by detecting and classifying road signs
- In this project, we aim to build a system that recognizes traffic signs using machine learning techniques

DATASET

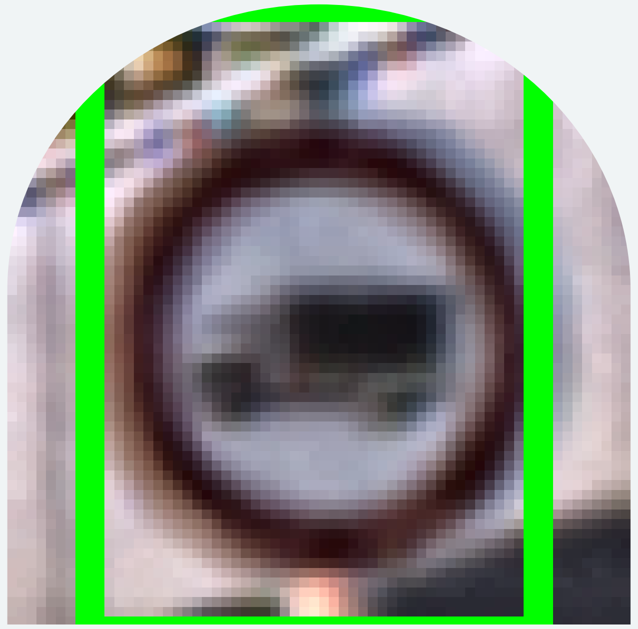
- Dataset used: GTSRB
- Contains 43 traffic sign classes
- Total images: 8,600
- Data split: 70% train, 15% validation, 15% test
- Dataset is balanced with equal images per class

PREPROCESSING

- Removed corrupted and duplicate images
- Balanced dataset (200 images per class)
- Resized all images to 64×64
- Split dataset into train / validation / test

DETECTION

- Used color-based detection (Red & Blue)
- Extracted bounding boxes for traffic signs
- If no detection → used full image
- Focused on Region of Interest (ROI)



FEATURE EXTRACTION

- Color Features:
  - RGB histogram
  - HSV histogram
- Shape Features:
  - Canny edges
  - HOG (Histogram of Oriented Gradients)
- Combined into one feature vector

MODEL TRAINING

- Two models were used:
  - SVM (Support Vector Machine)
  - CNN (Convolutional Neural Network)
- Both trained on extracted features
- Data normalized before training

SVM

- Trained on extracted feature vectors
- Advantages:
  - Works well with structured data
  - Effective for high-dimensional features
  - Fast and stable

- Accuracy: 91.04%
- Precision: 91.75%
- Recall: 91.04%
- F1 Score: 91.22%

Key Insight:

- Best performing model in this project
- Strong performance across all classes

CNN

- Architecture: Conv1D + Dense layers
- Input: feature vectors (not raw images)
- Advantages:
  - Can learn complex patterns
  - Handles non-linear relationships

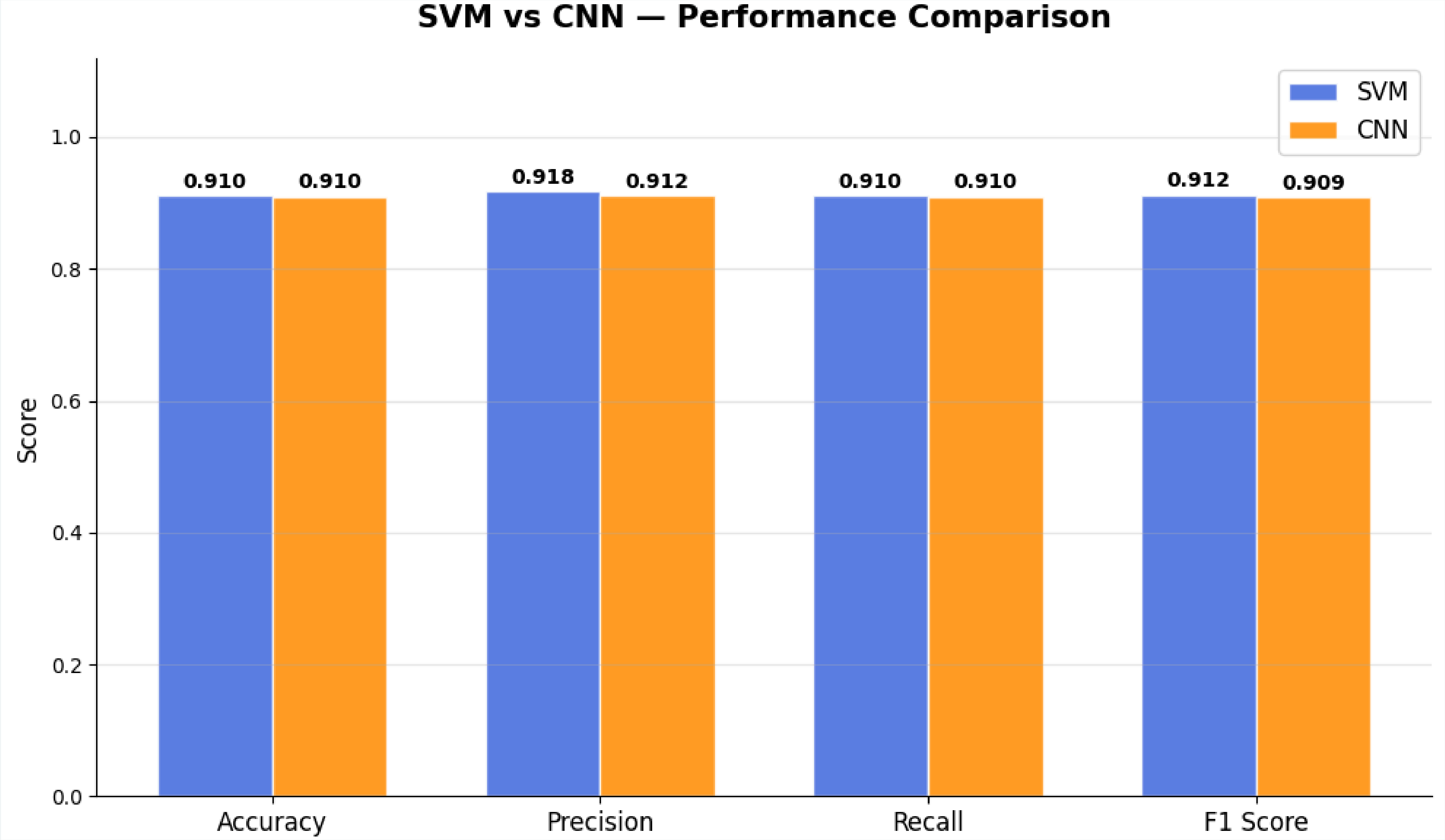
- Accuracy: 90.97%
- Precision: 91.16%
- Recall: 90.97%
- F1 Score: 90.94%

Key Insight:

- Performance very close to SVM
- Slightly lower results across metrics

SVM VS CNN

- SVM performed better than CNN
- Both models used same features
- Small performance difference



CONCLUSION

1. Built a traffic sign recognition system
2. Used feature extraction + classification
3. SVM achieved best performance (~91%)
4. System is accurate and reliable

THANK YOU